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MATHEMATICS - 2009

Objective Questions: (Q.1-Q.15)

- (1.) Let V be the vector space of all 6×6 real matrices over the field $\mathbb{R}$. Then the dimension of the subspace of V consisting of all symmetric matrices is
- (a.) 15
 - (b.) 18
 - (c.) 21
 - (d.) 35
- (2.) Let T be the ring of all functions from $\mathbb{R}$ to $\mathbb{R}$ under point-wise addition and multiplication. Let $I = \{f : \mathbb{R} \rightarrow \mathbb{R} \mid f \text{ is a bounded function}\}$, $J = \{f : \mathbb{R} \rightarrow \mathbb{R} \mid f(3) = 0\}$. Then
- (a.) J is an ideal of R but I is not an ideal of R
 - (b.) I is an ideal of R but J is not an ideal of R
 - (c.) Both I and J are ideal of R
 - (d.) Neither I nor J is an ideal of R
- (3.) Which of the following sequence of functions is uniformly convergent on $(0, 1)$?

(a.) x^n

(b.) $\frac{n}{nx+1}$

(c.) $\frac{x}{nx+1}$

(d.) $\frac{1}{nx+1}$

- (4.) Let $T: \mathbb{R}^4 \rightarrow \mathbb{R}^4$ be a linear transformation satisfying $T^3 + 3T^2 = 4I$, where I is the identity transformation. Then the linear transformation $S = T^4 + 3T^3 - 4I$ is

- (a.) One-one but not onto
- (b.) Onto but not one-one
- (c.) Invertible
- (d.) Non-invertible

- (5.) The number of all subgroups of the group $(\mathbb{Z}_{60}, +)$ of integers modulo 60 is

- (a.) 2
- (b.) 10
- (c.) 12
- (d.) 60

- (6.) Let $a_n = \begin{cases} \frac{1}{3^n}, & \text{if } n \text{ is prime} \\ \frac{1}{4^n}, & \text{if } n \text{ is not a prime} \end{cases}$

Then the radius of convergence of the power series $\sum_{n=1}^{\infty} a_n x^n$ is

(a.) 4

(b.) 3

(c.) $\frac{1}{3}$

(d.) $\frac{1}{4}$

(7.) The set of all limit points of the sequence

$$1, \frac{1}{2}, \frac{1}{4}, \frac{3}{4}, \frac{1}{8}, \frac{3}{8}, \frac{5}{8}, \frac{7}{8}, \frac{1}{16}, \frac{3}{16}, \frac{5}{16}, \frac{7}{16}, \frac{9}{16}, \dots$$
 is

(a.) $[0, 1]$

(b.) $(0, 1]$

(c.) The set of all rational numbers in $[0, 1]$

(d.) The set of all rational numbers in $[0, 1]$ of the form $\frac{m}{2^n}$ where m and n are integers.

(8.) Let $F: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function and

$a > 0$. Then the integral $\int_0^a \left[\int_0^x F(y) dy \right] dx$ equals

(a.) $\int_0^a yF(y) dy$

(b.) $\int_0^a (a - y)F(y) dy$

(c.) $\int_0^a (y-a)F(y)dy$

(d.) $\int_a^0 yF(y)dy$

(9.) The set of all positive values of a for which the

series $\sum_{n=1}^{\infty} \left(\frac{1}{n} - \tan^{-1} \left(\frac{1}{n} \right) \right)^a$ converges, is

(a.) $\left(0, \frac{1}{3} \right]$

(b.) $\left(0, \frac{1}{3} \right)$

(c.) $\left[\frac{1}{3}, \infty \right)$

(d.) $\left(\frac{1}{3}, \infty \right)$

(10.) Let a be a non-zero real number.

Then $\lim_{x \rightarrow a} \frac{1}{x^2 - a^2} \int_a^x \sin(t^2) dt$ equals

(a.) $\frac{1}{2a} \sin(a^2)$

(b.) $\frac{1}{2a} \cos(a^2)$

(c.) $-\frac{1}{2a} \sin(a^2)$

(d.) $-\frac{1}{2a}\cos(a^2)$

(11.) Let $T(x, y, z) = xy^2 + 2z - x^2z^2$ be the temperature at the point (x, y, z) . The unit vector in the direction in which the temperature decreases most rapidly at $(1, 0, -1)$ is

(a.) $-\frac{1}{\sqrt{5}}\hat{i} + \frac{2}{\sqrt{5}}\hat{k}$

(b.) $\frac{1}{\sqrt{5}}\hat{i} - \frac{2}{\sqrt{5}}\hat{k}$

(c.) $\frac{2}{\sqrt{14}}\hat{i} + \frac{3}{\sqrt{14}}\hat{j} + \frac{1}{\sqrt{14}}\hat{k}$

(d.) $-\left(\frac{2}{\sqrt{14}}\hat{i} + \frac{3}{\sqrt{14}}\hat{j} + \frac{1}{\sqrt{14}}\hat{k}\right)$

(12.) Consider the differential equation $2\cos(y^2)dx - xy\sin(y^2)dy = 0$. Then

(a.) e^x is an integrating factor

(b.) e^{-x} is an integrating factor

(c.) $3x$ is an integrating factor

(d.) x^3 is an integrating factor

(13.) Suppose $\vec{V} = p(x, y)\hat{i} + q(x, y)\hat{j}$ is a continuously differentiable vector field defined in a domain D

in $\mathbb{R}^2$. Which one of the following statements is NOT equivalent to the remaining ones?

(a.) There exists a function $\phi(x, y)$ such that

$$\frac{\partial \phi}{\partial x} = p(x, y) \quad \text{and} \quad \frac{\partial \phi}{\partial y} = q(x, y) \quad \text{for all}$$

$$(x, y) \in D$$

(b.) $\frac{\partial q}{\partial x} = \frac{\partial p}{\partial y}$ holds at all points of D

(c.) $\oint_C \vec{V} \cdot d\vec{r} = 0$ for every piecewise smooth closed curve C in D

(d.) The differential $p dx + q dy$ is exact in D

(14.) Let, $f, g: [-1, 1] \rightarrow \mathbb{R}$, $f(x) = x^3, g(x) = x^2|x|$.

Then

(a.) f and g are linearly independent on $[-1, 1]$

(b.) f and g are linearly dependent on $[-1, 1]$

(c.) $f(x)g'(x) - f'(x)g(x)$ is NOT identically zero on $[-1, 1]$

(d.) There exist continuous function $p(x)$ and $q(x)$ such that f and g satisfy $y'' + py' + qy = 0$ on $[-1, 1]$

- (15.) The value of c for which there exists a twice differentiable vector field $\vec{F}$ with $\text{curl } \vec{F} = 2x\hat{i} - 7y\hat{j} + cz\hat{k}$ is
- (a.) 0
 (b.) 2
 (c.) 5
 (d.) 7

Subjective Questions: (Q.16-Q.29)

- (16.) Container A contains 100 cc of milk and container B contains 100 cc of water. 5 cc of the liquid in A is transferred to B , the mixture is thoroughly stirred and 5 cc of the mixture in A is transferred back into A . Each such two-way transfer is called a dilution. Let a_n be the percentage of water in container A after n such dilutions, with the understanding that $a_0 = 0$.

(a.) Prove that $a_1 = \frac{100}{21}$ and that, in general,

$$a_n = \frac{100}{21} + \frac{19}{21}a_{n-1} \text{ for } n = 1, 2, 3, \dots$$

(b.) Using (a) prove that $a_n = 50 \left[1 - \left(\frac{19}{21} \right)^n \right]$ for

$n = 1, 2, 3, \dots$ Find $\lim_{n \rightarrow \infty} a_n$ and explain why the answer is intuitively obvious.

- (17.) (a.) Let $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ be a non-negative function. Assume that for every $m \in \mathbb{N}$, the series $\sum_{n=1}^{\infty} f(m, n)$ is convergent and has sum a_m and further that the series $\sum_{m=1}^{\infty} a_m$ is also convergent and has sum L . Prove that for every n , the series $\sum_{m=1}^{\infty} f(m, n)$ is convergent and if we denote its sum by b_n then the series $\sum_{n=1}^{\infty} b_n$ is also convergent and has sum L .

- (b.) Define $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ by

$$f(m, n) = \begin{cases} 0 & \text{if } n > m, \\ \frac{-1}{2^{m-n}} & \text{if } n < m, \\ 1 & \text{if } n = m. \end{cases}$$

Show that $\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} f(m, n) = 2$

and $\sum_{n=1}^{\infty} \sum_{m=1}^{\infty} f(m, n) = 0$.

- (18.) (a.) Evaluate $\iint_R \cos(\max\{x^3, y^{3/2}\}) dx dy$,

where $R = [0, 1] \times [0, 1]$.

(b.) Let $S = \sqrt{1} + \sqrt{2} + \sqrt{3} + \dots + \sqrt{10000}$ and

$$I = \int_0^{10000} \sqrt{x} \, dx. \text{ Show that } I \leq S \leq I + 100.$$

(19.) Let $D = \{(x, y) : x \geq 0, y \geq 0\}$.

Let $f(x, y) = (x^2 + y^2)e^{-x-y}$ for $(x, y) \in D$.

Prove that f attains its maximum on D at two

boundary points. Deduce that $\frac{x^2 + y^2}{4} \leq e^{x+y-2}$ for

all $x \geq 0, y \geq 0$.

(20.) (a.) Let $a_1, b_1, a_2, b_2 \in \mathbb{R}$. Show that the condition $a_2 b_1 > 0$ is sufficient but not necessary for the system

$$\frac{dx}{dt} = a_1 x + b_1 y,$$

$$\frac{dy}{dt} = a_2 x + b_2 y,$$

to have two linearly independent solutions

of the form $x = c_1 e^{\lambda_1 t}$, $y = d_1 e^{\lambda_1 t}$ and $x = c_2 e^{\lambda_2 t}$

, $y = d_2 e^{\lambda_2 t}$ with $c_1, d_1, c_2, d_2, \lambda_1, \lambda_2 \in \mathbb{R}$.

(b.) Show that the differential equation representing the family of all straight lines which have an intercept of constant length L between the coordinate axes is

$$x \frac{dy}{dx} - y = \frac{L \frac{dy}{dx}}{\sqrt{1 + \left(\frac{dy}{dx}\right)^2}}$$

(21.) Let $A, B, k > 0$. Solve the initial value problem

$$\frac{dy}{dx} - Ay + By^3 = 0, x > 0, y(0) = k.$$

Also show that

(a.) If $k < \sqrt{\frac{A}{B}}$, then the solution $y(x)$ is monotonically increasing on $(0, \infty)$ and

tends to $\sqrt{\frac{A}{B}}$ as $x \rightarrow \infty$.

(b.) If $k > \sqrt{\frac{A}{B}}$, then the solution $y(x)$ is monotonically decreasing on $(0, \infty)$ and

tends to $\sqrt{\frac{A}{B}}$ as $x \rightarrow \infty$.

(22.) (a.) Evaluate $\int_C (3y^2 + 2z^2) dx + (6x - 10z)y dy + (4xz - 5y^2) dz$ along the portion from $(1, 0, 1)$ to $(3, 4, 5)$ of the curve C , which is the intersection of the two surfaces $z^2 = x^2 + y^2$ and $z = y + 1$.

(b.) A particle moves counterclockwise along the curve $3x^2 + y^2 = 3$ from $(1, 0)$ to a point P , under the action of the force $\vec{F}(x, y) = \frac{x}{y}\hat{i} + \frac{y}{x}\hat{j}$. Prove that there are two possible locations of P such that the work done by $\vec{F}$ is 1.

(23.) Verify Stokes's theorem for the hemisphere $x^2 + y^2 + z^2 = 9$, $z \geq 0$ and the vector field $\vec{F} = (z^2 - y)\hat{i} + (x - 2yz)\hat{j} + (2xz - y^2)\hat{k}$.

(24.) (a.) Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be the linear transformation defined by $T(x, y, z) = (x + 2y, x - z)$.

Let $N(T)$ be the null space of T and

$$W = \{\vec{v} \in \mathbb{R}^3 \mid \vec{v} \cdot \vec{u} = 0, \text{ for all } \vec{u} \in N(T)\}.$$

Find a linear transformation $S : \mathbb{R}^2 \rightarrow W$ such that $TS = I$, where I is the identity transformation on $\mathbb{R}^2$.

(b.) Suppose A is a real square matrix of odd order such that $A + A^T = 0$. Prove that A is singular.

(25.) (a.) Find all pairs (a, b) of real numbers for which the system of equations

$$x + 3y = 1$$

$$4x + ay + z = 0$$

$$2x + 3z = b$$

- has (i) a unique solution,
 (ii) infinitely many solutions,
 (iii) no solution.

- (b.) Let A be an $n \times n$ matrix such that $A^n = 0$ and $A^{n-1} \neq 0$. Show that there exists a vector $v \in \mathbb{R}^n$ such that $\{v, Av, \dots, A^{n-1}v\}$ forms a basis for $\mathbb{R}^n$.

- (26.)** (a.) In which of the following pairs are the two groups isomorphic to each other? Justify your answers.

- (i) $\mathbb{R}/\mathbb{Z}$ and S^1 , where $\mathbb{R}$ is the additive group of real numbers and $S^1 = \{z \in \mathbb{C} : |z| = 1\}$ under complex multiplication.

- (ii) $(\mathbb{Z}, +)$ and $(\mathbb{Q}, +)$.

- (b.) Prove or disprove that if G is a finite abelian of order n , and k is a positive integer which divides n , then G has at most one subgroup of order k .

- (27.)** Let I and J be ideals of a ring R . Let IJ be the

set of all possible sums $\sum_{i=1}^n a_i b_i$, where $a_i \in I$,

$b_i \in J$ for $i = 1, 2, \dots, n$ and $n \in \mathbb{N}$.

(a.) Prove that IJ is an ideal of R and $IJ \subseteq I \cap J$.

(b.) Is it true that $IJ = I \cap J$? Justify your answer.

(28.) A sequence $\{f_n\}$ of functions defined on an interval I is said to be uniformly bounded on I if there exists some M such that $|f_n(x)| \leq M$ for all $x \in I$ and for all $n \in \mathbb{N}$.

(a.) Prove that if a sequence of function $\{f_n\}$ converges to a function f on I and $\{f_n\}$ is uniformly bounded on I , then f is bounded on I .

(b.) Suppose the sequences $\{f_n\}$ and $\{g_n\}$ of functions converge uniformly to f and g respectively on I and both are uniformly bounded on I . Prove that the product sequence $\{f_n g_n\}$ converges to fg uniformly on I . Show by an example that this may fail if only one of $\{f_n\}$ and $\{g_n\}$ is uniformly bounded on I .

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(29.) (a.) Prove that if f is a real-valued function which is uniformly continuous on an interval (a, b) , then f is bounded on (a, b) .

(b.) Let f be a differentiable function on an interval (a, b) . Assume that f' is bounded on (a, b) . Prove that f is uniformly continuous on (a, b) .



Rajendra Dubey (Known as Dubey Sir): He has experience of almost three decades of teaching mathematics. During his journey, he has guided IIT-JEE, IAS, NET, GATE, IIT-JAM aspirants etc. Apart from producing top rankers in the various exams, he is widely known for making mathematics interesting and fascinating. He is founder & director of DIPS ACADEMY and chairman of Dubey's Group of Companies.



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